

Aishwarya Jauhari Sharma

APPLIED AI ENGINEER

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Eligible to work in Germany, no sponsorship required

PROFILE

Applied AI engineer who builds language-model systems and puts them into production. Recent work is retrieval-augmented generation end to end: document parsing, chunking, embeddings and vector search over PostgreSQL with pgvector, with quantised local inference for privacy-sensitive data. Four years of full-time experience across applied research at Fraunhofer FHR, shipping an MLOps pipeline on FastAPI and Cloud Run, and consulting at Accenture delivering predictive models directly to US and Canadian clients. Comfortable owning a system from the business question to the deployed service.

CORE SKILLS

Generative AI and Machine Learning: LLM applications, retrieval-augmented generation, document parsing and chunking, embeddings and vector search, local and quantised inference, prompt design, deep learning, NLP, computer vision, time-series forecasting

Data Engineering and MLOps: MLflow, Prefect, Optuna, Docker, CI/CD with GitHub Actions, FastAPI, GCP Cloud Run, MongoDB, PostgreSQL, pgvector

Python and Frameworks: PyTorch, scikit-learn, OpenCV, HuggingFace, LlamaIndex, XGBoost, LightGBM, pandas, NumPy, statsmodels

Data Science: Statistical analysis, A/B testing, feature engineering, predictive modelling, time-series forecasting, customer segmentation, churn and risk modelling

Analytics and BI: SQL (PostgreSQL, MySQL, Redshift), query optimisation, Tableau, dashboard design, KPI monitoring, Matplotlib, Plotly, Excel

EXPERIENCE

Career Break: Maternity and Parenting

Sep 2024 – Present

Düsseldorf, Germany

- Maternity leave and full-time childcare. Kept technical skills current through self-driven projects and continued German language training (B1).

Applied Researcher, Machine Learning

Aug 2022 – Aug 2024

Fraunhofer FHR, Wachtberg, Germany

- Built a retrieval-augmented generation engine with document parsing so the team could query a corpus of research papers in natural language.
- Engineered a production-grade MLOps pipeline for SAR image change detection using FastAPI and Google Cloud Run.
- Developed deep learning models for Synthetic Aperture Radar (SAR) image classification, reducing manual target detection effort by 22% in defence applications.

Stack: PyTorch · OpenCV · CNNs · MLflow · Optuna · Docker · GitHub Actions · FastAPI · Google Cloud Run

Data Analyst (Working Student)

Nov 2020 – Jun 2022

mobile.de, Berlin, Germany

- Built machine learning models for user drop-off prediction by analysing user journey data with cross-functional teams, achieving a 12% lift in lead conversions.
- Optimised slow-running SQL queries and database performance, cutting daily marketing report generation from 8 minutes to 2 minutes.

Stack: Python · scikit-learn · pandas · NumPy · statsmodels · XGBoost · PostgreSQL · Tableau

Business Consultant

Nov 2016 – Dec 2018

Accenture, Bangalore, India

- Developed predictive models for a US mortgage client's risk assessment and customer segmentation (k-means), achieving 4% portfolio growth.
- Built product performance analytics dashboards tracking inventory turnover across a Canadian retail client's 200+ categories, reducing manual analysis time.

Stack: scikit-learn · Matplotlib · Redshift · MySQL · Tableau

SELECTED PROJECTS

5 end-to-end projects built during the career break. Code and write-ups at aishwaryai7.github.io.

DocTract, Local-First Document RAG Assistant

- Built a retrieval-augmented generation pipeline running entirely on the local machine: PyMuPDF parsing, sentence-boundary chunking, HuggingFace embeddings and PostgreSQL with pgvector for semantic retrieval.
- Ran generation on quantised Mistral 7B Instruct (Q4_K_M) through llama.cpp so documents never leave the host, with chunk size, top-k and search mode exposed for per-document tuning.

Stack: LlamaIndex · PyMuPDF · HuggingFace · PostgreSQL · pgvector · llama.cpp · Streamlit

Doc Parser, PDF Extraction and Table Question Answering

- Extracted text, images, tables and metadata from PDFs with PyMuPDF and converted them into structured JSON and Markdown for downstream use.
- Added natural-language question answering over the extracted tables using the OpenAI API.

Stack: Python · PyMuPDF · OpenAI API · Streamlit

MLOps Auto-Retraining Pipeline, Customer Churn

- Trained and compared logistic regression, random forest and gradient boosting in one scikit-learn preprocessing pipeline, logging every run to MLflow and registering the winner in the Model Registry with staging and production stages.
- Selected model scores 0.759 ROC-AUC on a held-out test set of 1,050 customers, behind a quality gate that blocks promotion below threshold.
- Served predictions from a FastAPI service with health, single and batch endpoints, containerised with Docker and deployed to Cloud Run through GitHub Actions.

Stack: Python · scikit-learn · MLflow · FastAPI · Docker · GitHub Actions · GCP Cloud Run

Also built: Energy Price Forecasting, European Electricity Markets (Python, XGBoost, LightGBM, statsmodels, MLflow, FastAPI, Streamlit, Docker, GCP). Speech2Text, Audio and Video Transcription (HuggingFace Transformers, PyTorch, torchaudio, Streamlit).

EDUCATION

M.Sc. Digital Engineering

Apr 2019 – Oct 2022

Otto-von-Guericke University, Magdeburg, Germany

Coursework: machine learning and data science, deep learning, business analytics, databases. Thesis: a deep learning framework for image segmentation, reaching 93% accuracy in automated body composition analysis of abdominal CT scans.

B.E. Instrumentation Technology

Sep 2012 – Aug 2016

Visvesvaraya Technological University, Bangalore, India

Coursework: microprocessors, C++, neural networks, logic design.

PUBLICATIONS (PUBLISHED AS AISHWARYA JAUHARI)

- Chatterjee, S., Sciarra, A., Dünwald, M., Tummala, P., Agrawal, S. K., Jauhari, A., Kalra, A., Oeltze-Jafra, S., Speck, O., Nürnberger, A. (2022). StRegA: Unsupervised anomaly detection in brain MRIs using a compact context-encoding variational autoencoder. *Computers in Biology and Medicine*, 149, 106093.
- Ataide, E. J. G., Agrawal, S., Jauhari, A., Boese, A., Illanes, A., Schenke, S., Kreissl, M. C., Friebe, M. (2021). Comparison of Deep Learning Algorithms for Semantic Segmentation of Ultrasound Thyroid Nodules. *Current Directions in Biomedical Engineering*, 7(2), 879–882.

CERTIFICATIONS

Deep Learning with PyTorch, IBM · Agile Foundations, LinkedIn Learning · University Defence Research Collaboration Summer School, University of Edinburgh

LANGUAGES

English (C1, business fluent) · German (B1, conversational, in progress)